PROBABILITY DISTRIBUTIONS

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 11

5-MINUTE REVISION

Source: Chapter 11 of the textbook.

1. QUICK REVISION · 1

- Probability distribution → gives possible values of a random variable and their probabilities.
- Discrete probability distribution → random variable takes countable values.
- Continuous probability distribution → random variable takes continuous values.
- Expected value → long-run average value of a random variable.
- Probability mass function → used for discrete random variables.

2. QUICK REVISION · 2

- Probability density function → used for continuous random variables.
- For a p.m.f., $p(x) \geq 0$.
- For a p.m.f., $\sum p(x) = 1$.
- Bernoulli distribution → exactly two possible outcomes, success and failure.
- Bernoulli parameter → p, the probability of success.

3. QUICK REVISION · 3

- Bernoulli mean → p.
- Bernoulli variance → $p(1-p)$.
- Binomial distribution → repeated Bernoulli trials.
- Binomial requirements → fixed number of trials, independent trials, and constant probability of success.
- Binomial parameters → n and p.

4. QUICK REVISION · 4

- Binomial failure probability → $q = 1 - p$.
- Binomial mean → np.
- Binomial variance → npq.
- In fitting a binomial distribution, $p = \frac{\bar{x}}{n}$.
- Expected frequency → $N \times p(x)$.

5. QUICK REVISION · 5

- Poisson distribution → models counts under appropriate rate and independence assumptions; it also approximates rare binomial successes.
- Poisson parameter → λ.
- Poisson mean → λ.
- Poisson variance → λ.
- In fitting a Poisson distribution, $\lambda = \bar{x}$.

6. QUICK REVISION · 6

- Normal distribution → continuous, bell-shaped and symmetric.
- Normal distribution parameters → μ and σ.
- Mean = median = mode in a normal distribution.
- Normal distribution is symmetric about its mean.
- Total area under the normal curve → 1.

7. QUICK REVISION · 7

- Standard normal distribution → mean 0 and standard deviation 1.
- Z-score → number of standard deviations an observation is from the mean.
- Standardization → subtract the mean and divide by the standard deviation.
- About 68.26% of observations lie within $\mu \pm \sigma$.
- About 95% of observations lie within $\mu \pm 1.96\sigma$.

8. QUICK REVISION · 8

- About 95.44% of observations lie within $\mu \pm 2\sigma$.
- About 99% of observations lie within $\mu \pm 2.58\sigma$.
- About 99.73% of observations lie within $\mu \pm 3\sigma$.
- Normal distribution has skewness 0.
- Normal distribution has kurtosis 3.

9. QUICK REVISION · 9

- All odd-order central moments are zero for the normal distribution.
- The total area under the normal curve represents total probability.
- The tails of the normal distribution are asymptotic to the horizontal axis.
- A linear combination of independent normal variables is also normally distributed.
- The Central Limit Theorem is closely associated with the normal distribution.

10. QUICK REVISION · 10

- Z-table gives cumulative probability to the left of a specified Z-score.
- Probability above a Z-score is obtained by subtracting the cumulative probability from 1.
- Probability between two Z-scores is obtained by subtracting the smaller cumulative probability from the larger one.

11. IMPORTANT FORMULAE · 1

Probability mass function for equally likely elementary outcomes:

$$P(X = x) = \frac{\text{Number of outcomes giving } X = x}{\text{Total number of elementary outcomes}}$$

Expected value of a discrete random variable:

$$E(X) = \sum_{x} x p(x)$$

Expected value of a continuous random variable:

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$

12. IMPORTANT FORMULAE · 2

Probability mass function:

$$p(x) = P(X = x)$$

Bernoulli probability mass function:

$$p(x) = p^x (1-p)^{1-x}, \quad x = 0, 1$$

Bernoulli mean:

$$E(X) = p$$

13. IMPORTANT FORMULAE · 3

Bernoulli variance:

$$Var(X) = p(1-p)$$

Binomial probability mass function:

$$p(x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \ldots, n$$

Combination form of binomial probability:

$$p(x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

14. IMPORTANT FORMULAE · 4

Probability of failure in binomial distribution:

$$q = 1 - p$$

Binomial mean:

$$E(X) = np$$

Binomial variance:

$$Var(X) = npq$$

15. IMPORTANT FORMULAE · 5

Estimation of binomial probability of success:

$$p = \frac{\bar{x}}{n}$$

Expected frequency in fitting a binomial distribution:

$$\text{Expected frequency} = N \times p(x)$$

Poisson probability mass function:

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

16. IMPORTANT FORMULAE · 6

Poisson mean:

$$E(X) = \lambda$$

Poisson variance:

$$Var(X) = \lambda$$

Estimation of Poisson parameter:

$$\lambda = \bar{x}$$

17. IMPORTANT FORMULAE · 7

Expected frequency in fitting a Poisson distribution:

$$\text{Expected frequency} = N \times p(x)$$

Normal probability density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < +\infty$$

Standard normal transformation:

$$z = \frac{X - \mu}{\sigma}$$

18. IMPORTANT FORMULAE · 8

Normal distribution notation:

$$X \sim N(\mu, \sigma^2)$$

Standard normal distribution notation:

$$X \sim N(0, 1)$$

Quartile deviation for a normal distribution:

$$Q.D. \approx 0.67449\sigma$$

19. IMPORTANT FORMULAE · 9

Mean deviation for a normal distribution:

$$M.D. = \sigma\sqrt{2/\pi} \approx 0.79788\sigma$$

20. WORKED EXAMPLE FROM THE CHAPTER

What is the z-score of a value of 27, given a set mean of 24, and a standard deviation of 2?

Solution

To find the z-score we need to divide the difference between the value, 27, and the mean, 24, by the standard deviation of the set, 2. $z = \frac{27-24}{2} = \frac{3}{2} = 1.5$. This indicates that 27 is +1.5 standard deviations above the mean.

21. SELF-CHECK FROM THE CHAPTER

1. The probability function of a discrete random variable is called a probability _____ function. **[Mass]**
2. The sum of all probabilities in a discrete distribution is _____. **[One]**
3. The long-run average value of a random variable is its _____ value. **[Expected]**
4. A single trial with outcomes coded 0 and 1 follows a _____ distribution. **[Bernoulli]**